HepaTwin

Automated liver-fibrosis screening and triage for primary care in India

From a routine CBC + LFT to a referral decision in seconds.

LIVE: hepatwin.vercel.app



Two outcomes of the same four routine lab values (synthetic demo cohort)

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Missed where patients first appear

Fibrosis is silent, and the first doctor rarely has a way to flag it

50M+

Indians live with chronic hepatitis B or C

1 in 3

urban adults has fatty liver disease (NAFLD)

Silent

no pain until cirrhosis: ascites, varices, encephalopathy

Routine labs

CBC and LFT are already ordered, but fibrosis risk is rarely calculated

Access gap

elastography and hepatology are concentrated in metro centres

No triage

Tier-2/3 clinics lack a simple rule for who needs referral

The gap is not data. It is a cheap, automatic way to act on data already in the report.

Built for primary care, Tier- 2/3 clinics

One engine, three people who benefit

GP

Primary care physician

Enter four values from a routine report. Get a risk tier and the next step in seconds.

Lab

Clinic or diagnostic lab

Auto-flag fibrosis risk on every CBC + LFT report, with no new equipment.

Hep

Referral hepatologist

Receives prioritised, explained referrals instead of undifferentiated ones.

Automated screening and triage. Decision support, never a diagnosis.

Four routine values in, a decision out

Validated, published scores first. Machine learning is a labelled second opinion

1 Inputs

Age, AST, ALT, platelets from a standard CBC + LFT. Optional: albumin, BMI, diabetes.

2 Validated scores

FIB-4 and APRI, with formulas visible. NAFLD Fibrosis Score when extra inputs exist.

3 Triage

Low, indeterminate or high tier with a plain-language next step.

$$\text{FIB-4} = (\text{Age} \times \text{AST}) \div (\text{Platelets} \times \sqrt{\text{ALT}})$$

$$\text{APRI} = (\text{AST} \div \text{upper limit of normal}) \div \text{Platelets} \times 100$$

Platelets in 10^9/L. Published, widely used formulas: no black box in the first step.

Three tiers, three clear next steps

Published FIB-4 thresholds drive the tier. Local protocol always governs

LOW

FIB-4 below 1.3

Manage in clinic. Lifestyle and metabolic counselling, periodic repeat labs.

INDETERMINATE

FIB-4 1.3 to 2.67

Second-line test: elastography or ELF at the nearest centre. Recheck labs.

HIGH

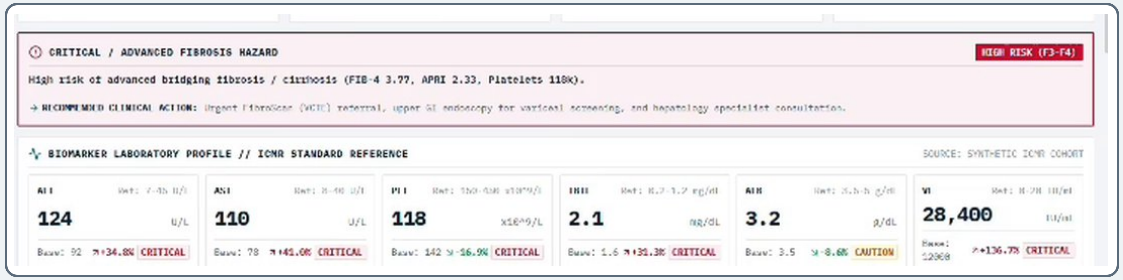
FIB-4 above 2.67

Refer to hepatology. Consider elastography and variceal screening.

Age 65 and over uses the lower 2.0 cutoff. APRI is shown alongside for concordance.

What the primary care doctor sees

One screen: tier, reason, recommended action, and the labs behind it





Live app screens, synthetic demo cohort

Tier and action

Plain-language next step in the same panel

Labs in context

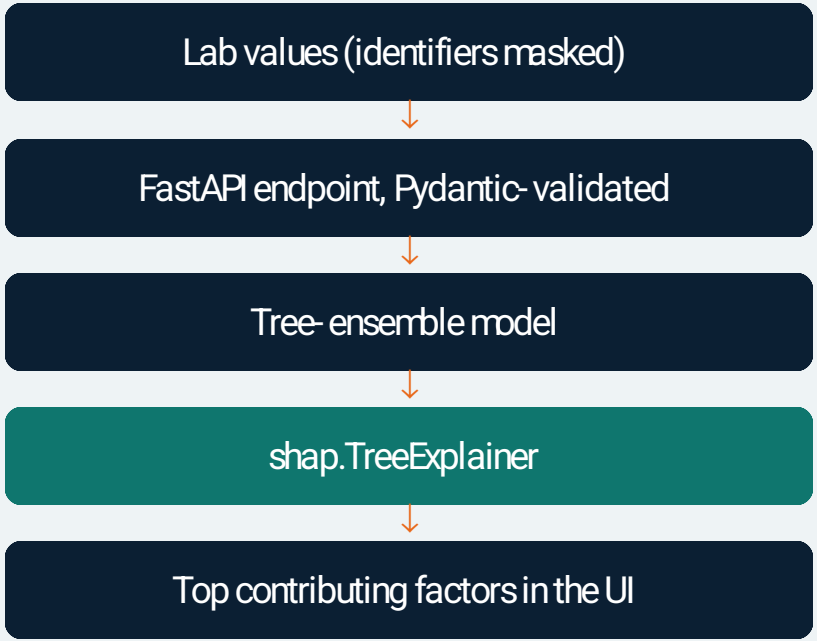
Each value against its reference range and baseline

Visible formula

FIB-4 and APRI one hover away

Explainability from real SHAP values

Per-patient attributions computed server-side, not hard-coded rules

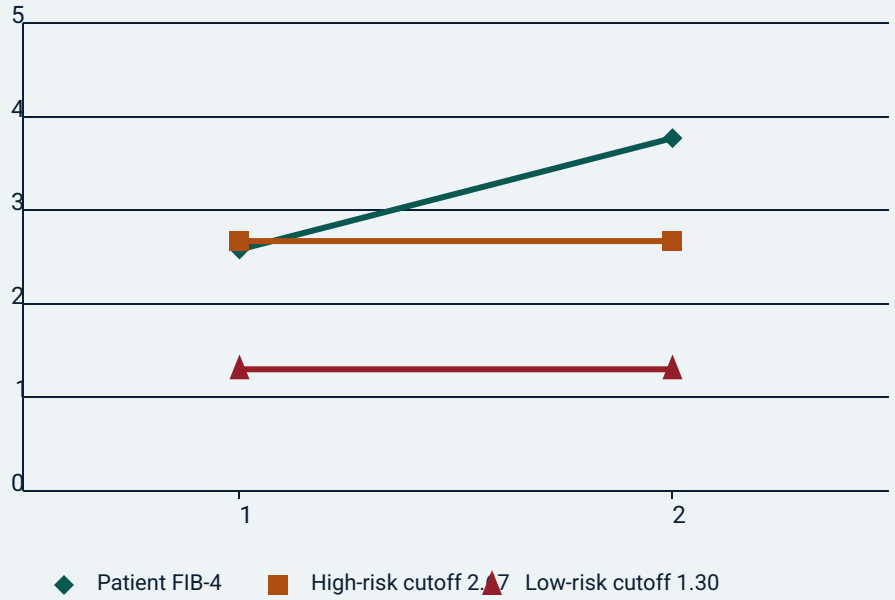


Why this is credible

- Transparent first: FIB-4 and APRI are the primary, auditable score
- SHAP explains the ML second opinion for each individual patient
- Same inputs always give the same attributions
- Attributions explain the model, not biology: no causal claims

Trends fromthe patient's own labs

Serial FIB-4 fromdated lab panels, not hand-set drift sliders



Fromthe app's own data

- Baseline AST 78, ALT 92, platelets 142: FIB-4 2.58
- Current AST 110, ALT 124, platelets 118: FIB-4 3.77
- Crosses the 2.67 high-risk cutoff: escalate
- Next: fit validated progression models on real serial cohorts

Validation: how we earn trust

We make no accuracy claim today. Here is the plan that would justify one

1 Reference standard

Transient elastography (FibroScan) and biopsy where available, on real patients.

2 What we report

Sensitivity, specificity, PPV and NPV at the referral thresholds, plus calibration, against FIB-4 alone.

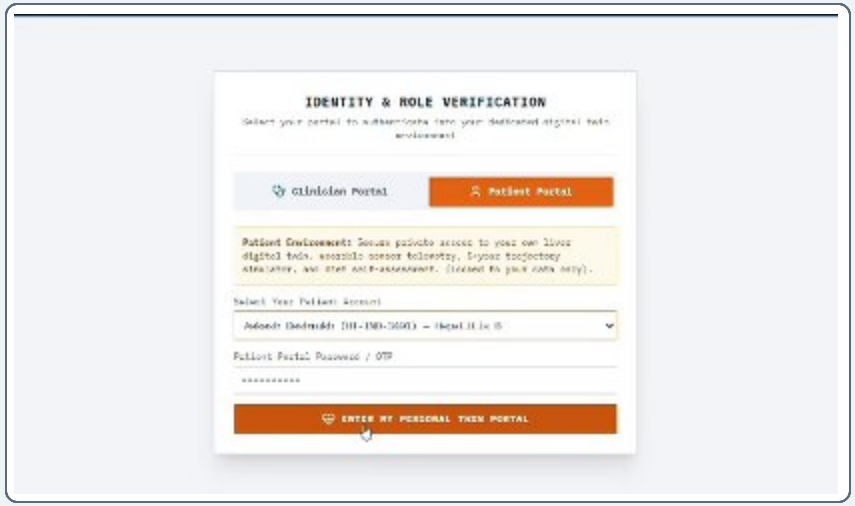
3 Where we test

Retrospective Indian clinic and lab data first, then a prospective pilot across more than one site.

Today's demo runs on a synthetic cohort to show the workflow. It is not clinical evidence.

Privacy and interoperability by design

Built to fit into real clinics and national health infrastructure



Role-based sign-in

DPDP 2023

Client-side masking turns names and IDs into hashes before transmission

ABDM and FHIR R4

ABHA IDs, FHIR fields and consent artifacts

Engineering checks

Vitest 11 of 11 passing, TypeScript strict with 0 errors

Pilot plan and what we will measure

Targets to measure in Tier-2/3 clinics, not claims we are making today

Phase 1
Retrospective validation on real labelled data

Phase 2
Shadow-mode pilot: flag risk, doctors decide as usual

Phase 3
Live triage in clinics and diagnostic labs

Coverage: share of routine CBC + LFT reports with a fibrosis risk computed

Referral quality: share of referred patients who truly need specialist care

Avoided referrals: low-tier patients safely managed in primary care

Time to specialist: days from high-tier flag to hepatology visit

Catch fibrosis at the first blood test

Low- cost screening and triage for primary care and Tier- 2/3 clinics across India.

Routine labs
No new equipment, just CBC + LFT

Transparent
Published formulas plus real SHAP

Right referral
Elastography and hepatology for those who need them

Try it now: hepatwin.vercel.app

Code: github.com/Parthkala/Repository1

Decision support for screening and triage. Not an approved medical device. Demo data is synthetic.